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RESEARCH ARTICLE

Comprehension of irony in autistic children: The role of theory of mind and executive function

Yongning Song[1,2] | Ziyun Nie[1,2] | Jiatong Shan[3]

[1]Shanghai Key Laboratory of Brain Functional Genomics, Key Laboratory of Brain Functional Genomics Ministry of Education, School of Psychology and Cognitive Science, East China Normal University, Shanghai, China

[2]Shanghai Key Laboratory of Mental Health and Psychological Crisis Intervention, School of Psychology and Cognitive Science, East China Normal University, Shanghai, China

[3]Department of Arts and Science, NYU Shanghai University, Shanghai, China

Correspondence
Yongning Song, School of Psychology and Cognitive Science, East China Normal University, North Zhongshan Road 3663, Shanghai 200062, China.
Email: ynsong@psy.ecnu.edu.cn

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Abstract
Although previous studies have examined irony comprehension in autistic children and potential impact factors, the relationship between theory of mind (ToM), executive function (EF), symptoms of autism, and comprehension of irony in this population remains largely unknown. This study explored irony comprehension in autistic children and examined the roles of ToM and EF in linking autism symptoms to deficits in irony comprehension. Twenty autistic children were compared with 25 typically developing (TD) children in an irony story picture task, ToM task, and EF task. The results showed that autistic children had impaired comprehension of irony compared with TD children, and performance on ironic stories showed a significant moderate discriminatory effect in predicting autistic children. A ToM deficit has also been proposed for autistic children. Comprehension of irony was significantly correlated with second-order ToM (2nd ToM) but was not significantly correlated with any components of EF. Moreover, 2nd ToM can predict the level of irony comprehension and mediate the relationship between symptoms of autism and irony comprehension. Taken together, these findings suggest that irony comprehension may offer a potential cognitive marker for quantifying syndrome manifestations in autistic children, and 2nd ToM may provide insight into the theoretical mechanism underlying the deficit in irony comprehension in this population.

Lay Summary
This research shows that autistic children have specific cognitive deficits relating to understanding irony, meanwhile, second-order theory of mind may influence their comprehension of irony. This finding suggests that these deficits could act as a cognitive marker for autism spectrum disorder, allowing for earlier identification and treatment of autistic children.

KEYWORDS
autism spectrum disorder, executive functions, irony comprehension, theory of mind

INTRODUCTION

Irony is a rhetorical expression that indicates an intention opposite to its literal meaning. It is commonly used on social occasions in daily life and tends to involve complex emotions (Pexman et al., 2010). Through irony, speakers can emphasize the differences between expectations and reality (Gibbs Jr et al., 1994) and achieve certain social goals, especially by indirectly conveying a certain attitude (Dews et al., 1996). Thus, correct comprehension of irony is a prerequisite for better social interaction. Irony can be classified into two types: ironic criticism and ironic praise. Ironic criticism usually appears in the form of positive statements that imply an underlying negative connotation, generally with sarcasm. For example, a person waits a long time for someone who is late and says "You're very quick." Ironic praise refers to negative statements conveying positive implications. For example, you might tell a friend who has won a big prize, "It's dumb luck."

Autism spectrum disorder (ASD) is a neurodevelopmental disorder characterized by deficits in communication and social interactions (American Psychiatric

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Association [APA], 2013). The inability to understand nonliteral meanings was added to the diagnostic criteria for ASD in the DSM-5 (Kocsis, 2013). Previous research showed that autistic children can only interpret conversation literally and take less notice of the speaker's actual intention, which leads to deficits in social communication and interaction (Panzeri et al., 2022). Based on studies on impairments in nonliteral meaning comprehension in individuals with ASD, many researchers have begun to explore these individuals' ability to understand irony (Happé, 1993).

Theory of mind and irony comprehension in autistic children

The ability to infer others' mental states has been termed "theory of mind" (ToM; Wellman & Gelman, 1992). Researches have shown that autistic children present impairment in ToM (e.g., Baron-Cohen et al., 1985; Polyanskaya et al., 2022). Baron-Cohen et al. (1985) discovered deficits in ToM in autistic children using the classic false-belief task. ToM is often assessed using first- and second-order false-belief tasks (i.e., 1st ToM and 2nd ToM; Perner et al., 1987; Wimmer & Perner, 1983); 1st ToM refers to deducing another person's opinion about events, and 2nd ToM usually enables individuals to infer another person's ideas while considering their perceptions and attitudes toward that idea, that is, understanding nested mental states (Li et al., 2013; Perner & Wimmer, 1985).

Understanding irony requires the ability to infer that the speaker's expressed intention is contrary to the literal meaning of the speech. Previous studies have shown that, compared with typically developing (TD) children, autistic children have more difficulty understanding irony (e.g., Huang et al., 2015; Saban-Bezalel et al., 2019). Because irony comprehension not only involves literal meaning but also includes understanding the speaker's attitudes and hidden meanings (Happé, 1993; Zhang & Zhang, 2009), researchers have found that 2nd ToM is closely correlated with children's irony comprehension (Bosco & Gabbatore, 2017). Children with higher levels of 2nd ToM are also more capable of processing information, dealing with double-nested problems, and have higher accuracy in irony comprehension (Zhang & Zhang, 2009). Autistic individuals who have well-developed ToM can understand most figurative language, which is based on nonliteral meanings, including the most difficult type: irony (Bosco & Gabbatore, 2017; Li et al., 2013).

Executive function and irony comprehension in autistic children

Understanding irony requires the ability to alter literal and nonliteral meanings rapidly and flexibly and to inhibit understanding of irrelevant literal meanings in the context (Cordonier et al., 2020). The ability for high-order processes like planning and controlling goal-directed behavior has been termed "executive function" (EF; Miyake et al., 2000). Deficits in EF could explain children's impairment in irony comprehension to some extent. Many studies have demonstrated that inhibition and nonliteral language understanding are positively associated (Champagne & Joanette, 2004; Champagne-Lavau & Joanette, 2009; Channon & Crawford, 2010) and that flexibility and working memory also play a role in irony comprehension (Channon & Watts, 2003; Martin & McDonald, 2005; Zimmermann et al., 2011).

Several studies have indicated that executive dysfunction is a core feature in autistic children (Fossum et al., 2021; Verté et al., 2006), including deficits in working memory (e.g., Barnard et al., 2008), flexibility (e.g., Ozonoff et al., 2004), and behavioral perseverance (e.g., Prior & Hoffmann, 1990). EF is associated with two major symptoms of ASD: repetitive and restrictive behaviors and impaired social communication and interaction (Hill, 2004). In autistic children, findings have also revealed correlations between EF and several types of figurative language understanding whose expressions have intended meanings that are inconsistent with their literal meanings (e.g., metaphors and proverbs; Kasirer & Mashal, 2016; Mashal & Kasirer, 2011; Tzuriel & Groman, 2017). Additionally, when participants' EFs were matched, children performed better on irony comprehension tasks (Saban-Bezalel et al., 2019).

This study

This study investigated whether autistic children have an impaired ability to understand irony. It also aimed to examine the mechanism underlying deficits in irony comprehension by investigating the roles of autism symptoms, ToM, and EF in understanding irony. This study was guided by the following research questions.

1. How well can autistic children understand irony, and is the ability a potential behavioral marker for quantifying syndrome manifestation in autistic children? We hypothesized that autistic children in China are impaired in irony comprehension compared with TD children, and the prediction effect is significant in distinguishing autistic children through signal detection analysis and receiver operating characteristic (ROC) curves.
2. What are the relationships between autism symptoms, ToM, EF, and irony comprehension? We hypothesized that individuals with more autism symptoms would tend to exhibit lower levels of EF and ToM, and subsequently exhibit a lower level of irony comprehension. It is expected that autism symptoms will have negative effects on EF, ToM, and irony

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comprehension, whereas EF and ToM will have direct positive effects on irony comprehension. Specifically, EF and ToM mediate the relationship between autism symptoms and irony comprehension in autistic children.

METHODS

Participants

All participants were recruited online. Before the study, we confirmed that all of them had been diagnosed with HFA/ AS at Shanghai Pediatric Hospital or similar institutes in Shanghai. The same procedure we also used in our previous study (Liu et al., 2023). The children in the ASD group met the DSM-5 diagnostic criteria. We investigated the medical and developmental history, behaviors, and symptoms of the children in the ASD group in advance to exclude them from serious comorbidities such as attention-deficit hyperactivity disorder, epilepsy, or schizophrenia. Written informed consent was obtained from the participants and their parents before testing.

Before the formal experiment, all participants were tested using the Combined Raven's Test (CRT; Raven et al., 1992), high-functioning Autism Spectrum Screening Questionnaire (ASSQ; Ehlers et al., 1999), and Autism-spectrum Quotient (AQ; Baron-Cohen et al., 2001). Intelligence quotient (IQ) scores were calculated based on CRT scores and age. Children with intellectual disabilities (IQ < 85) were excluded. We also adopted a conventional classification procedure in which children with an AQ score ≥30 and an ASSQ score >13 were included (Ehlers et al., 1999).

Twenty-five participants with ASD (23 males and 2 females, with an average age of 13.37 years) were ultimately obtained. The control group was recruited from a primary school in Shanghai and included 25 TD children (16 males and 9 females, with an average age of 13.46 years). The control group had no history of mental illness or developmental disorders. There was no significant difference between the two groups in terms of average age and CRT score (t [47] = −0.164, $p = 0.870$; t [47] = −0.501, $p = 0.619$); however, there was a significant difference in AQ (t [47] = −12.000, $p < 0.001$). Using G*power software (Faul et al., 2007), a total sample size of around 50 participants was determined as eligible for an a priori power analysis with the prespecified significance level alpha of 0.05, a power level of 0.95, a medium effect size ($f = 0.25$) of repeated measures analyses of variance (ANOVAs; see also previous study with a comparable sample size, Song & Hakoda, 2018). Additional data on socioeconomic status and educational attainment levels were not recorded. Community members were not involved in the design of the study. The study was approved by the University Committee on Human Research Protection (HR 220-2019). Participants' demographic and symptom information is given in Table 1.

Measures

Irony tasks

We developed an irony story task based on tasks used in previous research (Zhang & Zhang, 2009). The stories occurred in four daily life contexts (one family and three schools). Four stories (literal praise, literal criticism, ironic praise, and ironic criticism) were designed for each context. Each story included the same two characters (i.e., a father and child), ending with a key sentence that the father says to the child (see Table 2). In literal stories, the literal meaning corresponds to the actual intention of the sentence, while in ironic stories, it is contrary to the actual intention. In the ironic praise task, the literal meaning is praise, whereas the metaphorical meaning is criticism. Conversely, in the ironic criticism task, the literal meaning was criticism, while the metaphorical meaning was praise.

In order to confirm the content validity of the task, materials were assessed through accuracy rate by 32 college students who were recruited to judge the type of stories. The result showed that the accuracy rates of literal praise and literal criticism were 99.22%, irony criticism was 98.44%, and irony praise was 96.88%, indicating a great validation of tasks.

When measuring irony comprehension in our study, story cartoons were explained to participants by an experimenter with no special expression, intonation, or prosody, etc. Given participants' age and language ability, we adopted this form to ensure participants' understanding of the main plots in stories. After each story, the children were asked to judge the speaker's emotions, intentions, and attitudes conveyed by the last key sentence in each story.

The instructions for this task are as follows: "Welcome to the test. You will hear four short stories. Please listen carefully and answer several questions related to them." Pictures of the story content were shown along with each story to facilitate understanding. The order of the stories was counterbalanced during the tests. After telling each story, the experimenter asked "Did you understand the story?" "What did Xiaohua (or Lanlan) do in the story?" and "What did the mother (or father) say to Xiaohua (or Lanlan)?" to check whether the children really remembered and understood the story. If the questions were answered correctly, the formal test was conducted; otherwise, the story was repeated. After each story, participants were required to judge the father's emotions, intentions, and attitudes conveyed by the key sentence that the father said to the child at the end of the story. If it was difficult for the participants to answer, pictures of three types of emotions (happy, neutral, and

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TABLE 1 Demographic and symptom variables of participants.

	ASD group	Control group	*t* (*df*)	*p*-Value
Sex	Male (23), female (2)	Male (16), female (9)	Chi square (1) = 5.71*	0.02
Average age (SD)	13.37 (2.08)	13.42 (1.53)	t (48) = −0.09	0.93
Age range	10.20–18.30	10.60–16.10		
Raven IQ test (SD)	112.00 (14.63)	110.2 (12.34)	t (48) = −0.47	0.64
AQ (SD)	27.00 (7.94)	4.25 (4.99)	t (48) = −11.98***	0.001

Abbreviations: ASD, autism spectrum disorder; AQ, Autism-spectrum Quotient; IQ, Intelligence quotient.
*$p < 0.05$; ***$p < 0.001$.

TABLE 2 An example of irony stories.

Story types and check questions	Scripts
Literal praise	Lanlan's father was lying in bed because of illness. Lanlan got some medicine from the drugstore, and her father gradually got better after taking the medicine. Her father told her: "You are such a good kid!"
Ironic praise	Lanlan's father was lying in bed because of illness. Lanlan got some medicine from the drugstore, and her father gradually got better after taking the medicine. Her father asked her: "How can you blow your money?"
Literal criticism	Lanlan's father was lying in bed because of illness. Lanlan did not use the money to buy medicine for her father; instead, she bought a toy car she liked. Her father asked her: "How could you blow the money?"
Ironic criticism	Lanlan's father was lying in bed because of illness. Lanlan did not use the money to buy medicine for her father; instead, she bought a toy car she liked. Her father told her: "You are such a good kid!"
Checked questions	Emotions: What was the father's emotion when he said this? Happy, sad, or no idea? Intentions: Do you think father was criticizing Lanlan or praising her? Criticize, praise, or no idea? Attitudes: Was father satisfied with Lanlan or not? Satisfied, dissatisfied, or no idea.

sad) were provided to help. After answering the questions for one story, the experimenter moved on to the next story. The irony test used 12 questions (3 questions in each test in 4 contexts) involving emotion, attitude, and intention dimensions, with one point for a correct answer and zero points for an unknown or incorrect answer. The accuracy rate was calculated by dividing the final score by the maximum score (i.e., 12 points).

ToM tasks

This study used 12 1st ToM and 2nd ToM tasks, designed based on Baron-Cohen (1996) and Perner and Wimmer (1985). The test–retest Kappa values for the original Tom tasks all fell between 0.40 and 0.80, indicating good reliability (Hughes et al., 2000). The 1st ToM tasks included two unexpected locations, two unexpected content tasks, the distinction between reality and imagination, and the inference of others' mental states. The 2nd ToM tasks asked participants to infer where character B would go (six unexpected location tasks) and how Character B felt from the perspective of Character A (six questions on emotion inference). Participants were presented with cartoons of the stories while they were narrated. At the end of each story, participants were asked a first- or second-order belief question (see Table 3).

An example of a 1st ToM task is presented in the following scenario. Mary and John play ball in the room. After some time, John put his ball into a box and played outside. After John left, Mary took the ball from the box and put it in a basket. Later, John returned to the room and wanted to play with his ball again. The false-belief question was, "Where would John look for his ball, in the box or the basket?"

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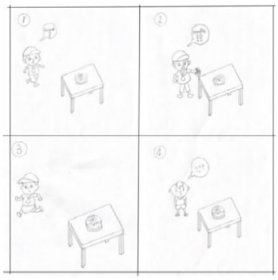
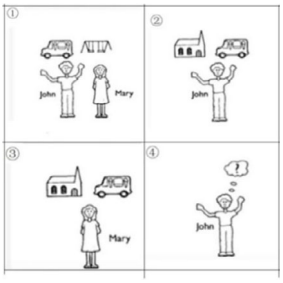
TABLE 3 An example of ToM task.

Story types		Scripts
1st ToM (emotion)		Xiao Li was once bitten by a dog. Today, while she was taking a leisurely walk outside, she heard barks and there was a small dog rushing up to her, wanting to play with her. How do you think Xiao Li would feel?
1st ToM (cognition)		Xiao Ming saw a biscuit tin on the table. When he opened it, he found that his mother had placed noodles instead of biscuits inside. He felt very disappointed and put it back. After a while, Xiao Ming's little brother came, and he also saw the cookie tin. What would the little brother think was inside the tin?
2nd ToM (emotion)		On the day of Xiao Hong's birthday, her dad bought a birthday cake and playfully pretended not to have bought one, teasing Xiao Hong. How does Xiao Hong's dad think Xiao Hong would feel after hearing this?
2nd ToM (cognition)		Mary and John saw an ice cream truck at the park. Mary ran back home to get money, and at that moment, John saw the truck drive toward the church. Mary unexpectedly found the truck near the church. Where will John wait for Mary?

Abbreviations: 1st ToM, first-order theory of mind; 2nd ToM, second-order theory of mind.

An example of a 2nd ToM is as follows: One day, Mary and John were in the park, and they met a man selling ice cream. Mary wanted to eat ice cream but had no money. The man who sold ice cream told Mary: "You can go home to get money, and I won't go until you come back." Mary went home to get some money. After some time, the seller wanted to go somewhere he could sell more. John asked the man, "Where will you go?" The seller said, "I will go to the church, where I can sell more." Then John went home. When he went past Mary's home, he asked Mary's mother if Mary was in, and Mary's mother said that Mary had gone to buy ice cream. The belief question was "Where does John think Mary has gone to buy ice cream, the park or the church?" The 2nd ToM task included 12 questions involving cognition and emotion dimensions, with scores ranging from 0 to 12.

EF tasks

The EF task is a battery of working memory measured by a spatial working memory task (SWM; Robbins et al., 1998), inhibitory control measured by a go/no-go task (Robbins et al., 1998), and cognitive flexibility measured by a dimensional change card sort test (DCCS, Zelazo, 2006). The SWM and go/no-go tasks were adapted from the classical neuropsychological task (i.e., Cambridge Neuropsychological Test Automated Battery, CANTAB; Robbins et al., 1994), which was

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TABLE 4 Results of descriptive statistics and group differences in performance on irony tasks, executive function (EF) tasks, and theory of mind (ToM) tasks (mean ± SD).

		Accuracy rate/score			
		ASD group ($N = 25$)	TD group ($N = 25$)	*t*-Value	*p*-Value
Irony comprehension	Literal praise	0.93 (0.15)	0.94 (0.11)	−0.18	0.92
	Literal criticism	0.85 (0.30)	0.97 (0.08)	−1.98	0.08
	Ironic praise	0.53 (0.27)	0.75 (0.24)	−3.91***	<0.001
	Ironic criticism	0.45 (0.39)	0.81 (0.24)	−3.02***	<0.001
EFs	Inhibition	38.12 (6.89)	42.67 (2.48)	−3.1**	<0.01
	Flexibility	8.36 (2.41)	8.71 (2.76)	−0.48	0.64
	Working memory	33.56 (10.47)	36.46 (7.03)	−1.2	0.26
ToM	1st ToM	6.40 (2.78)	9.08 (1.82)	−4.03***	<0.001
	2st ToM	6.32 (3.25)	9.40 (1.96)	−4.06***	<0.001

Abbreviations: 1st ToM, first-order theory of mind; 2nd ToM, second-order theory of mind; ASD, autism spectrum disorder; TD, typically developing.
**$p < 0.01$; ***$p < 0.001$.

wildly used to test the EF. It also has been validated in autistic people through a great amount of research (e.g., Cordova et al., 2020; Feczko et al., 2018; Steele et al., 2007; Toseeb et al., 2018). The DCCS was a well-designed task widely used to test cognitive flexibility, especially in children (Zelazo, 2006). Its validity was verified in autistic children as well (Demetriou et al., 2019; Gonzalez-Barrero & Nadig, 2019; Haebig et al., 2015).

The SWM task tests the ability to retain and manipulate spatial information in working memory. The children were asked to find a cartoon picture in several boxes. Once a box was opened, the children were not permitted to check it again. The number of pictures varied sequentially from three to seven. The children must remember the location of the box through elimination. The number of incorrect responses was recorded. The final score was calculated according to the following formula (50 incorrect responses): The numbers of correct and incorrect responses were recorded, and the task continued until the participant gave 50 incorrect responses. The number of correct responses was the final score; The higher the final score, the better the performance on the spatial working memory task.

The go/no-go task measured the ability to inhibit responses. This task used 38 brown and 12 pink bear pictures. Children were instructed to respond to go-stimuli (brown bear) quickly and accurately, but not to the no-go stimuli (pink bear). Correct responses to go-stimuli and correct inhibition to no-go stimuli comprise a total score, ranging from 0 to 50.

In the DCCS task, cards of different colors and shapes were displayed. Children were instructed to sort the cards using one dimension (e.g., color). The children then needed to switch from one dimension to another (e.g., shape). They were also required to sort cards with a black frame according to color. The sequence of the sorting rule was counterbalanced among the participants. To guarantee the correct understanding of the rules, four cards were used for practice before the experiment. One point was awarded for each correct classification, with a total maximum score of 12.

Procedure

The experiment was conducted in a quiet room, and all participants performed three cognitive tests (i.e., irony stories, ToM task, and EF task), which took ∼30 min. Participants received small gifts (i.e., notebooks) that cost about five Chinese Yuan (∼USD 0.8) for participating. This study was approved by the Ethics Committee of East China Normal University.

Data analysis

Four sets of analyses were performed in this study. First, descriptive statistics and group differences in performance on the irony, EF, and ToM tasks were analyzed. Second, a signal detection analysis was conducted to ascertain the predictive effect of performance on different irony tasks in distinguishing between autistic children and TD children. Third, Pearson correlation analysis was conducted to reveal the relationship between irony comprehension, autism symptoms, and performance on ToM and EF tasks. Finally, regression and mediation analyses using SPSS 24.0 were conducted to test the relationships between ToM, EF, autism symptoms, and irony comprehension. Regression assumes that variables have normal distributions and non-normally distributed variables (highly skewed or kurtotic variables, or variables with substantial outliers) can distort relationships and significance tests (Osborne & Waters, 2002). Variables with sample size usually do not meet this assumption. In

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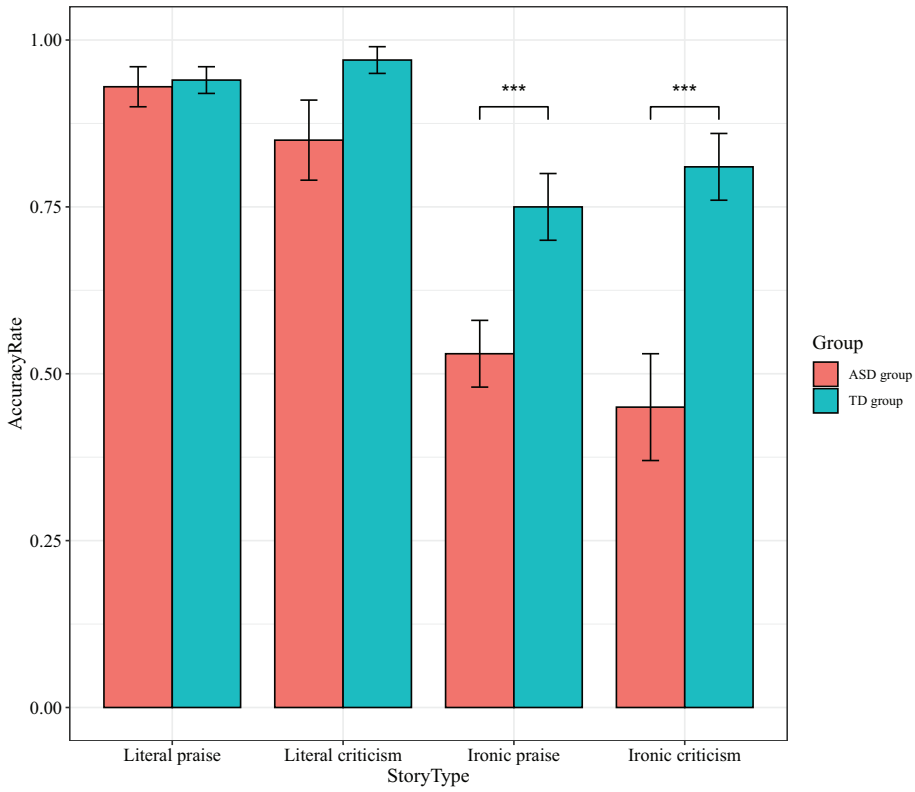

FIGURE 1 The accuracy ratio of four irony tasks in typically developing (TD) and autism spectrum disorder (ASD) group. The irony comprehension of TD group is significantly better than the irony comprehension of ASD group for both ironic criticism and ironic praise ($p < 0.001$); There were no significant group differences in both literal criticism and literal praise. Error bars: SEs. *** $p < 0.001$.

TABLE 5 Receiver operating characteristics (ROC) with area under the curve (AUC), optimal cutoff scores, sensitivity, and specificity for ironic criticism story, ironic praise story, literal criticism story, and literal praise story.

	Literal praise	Literal criticism	Ironic praise	Ironic criticism
ROC–AUC [95% CI]	0.409 [0.38–0.65]	0.52 [0.29–0.53]	0.72 [0.58–0.86]	0.76 [0.62–0.89]
Cutoff (sensitivity/specificity)	10.5 (0.84/0.20) 11.5 (0.72/0.32)	—	5.5 (0.44/0.92)	8.5 (0.64/0.84) 7.0 (0.60/0.88)

addition, using G*power software (Faul et al., 2009), a total sample of at least 40 participants was estimated in the prior power analysis, with the prespecified significance level alpha of 0.05, a power level of 0.95, and a large effect size ($r = 0.5$, $f^2 = 0.35$) of correlation and linear multiple regression in the current research. Given the total number of participants in our research is 50, we treated the entire sample as one group to analyze correlation and linear multiple regression.

RESULTS

Descriptive statistics and group differences in performance on irony, EF, and ToM tasks

Descriptive data of participants' performances in all tasks are presented in Table 4. We performed a two-way repeated ANOVA with the irony comprehension score for the group type (ASD and TD groups) as a between-subjects factor, and the story type (literal criticism, literal praise, ironic criticism, and ironic praise) as the within-subject factor. The results show that the main effects of group (F [1,48] = 25.55, $p < 0.001$, $\eta^2 = 0.331$,) and story type (F [3,144] = 22.70, $p < 0.001$, $\eta^2 = 0.347$) were significant. The interaction between group type and story type was also significant (F [3,144] = 4.76, $p < 0.01$, $\eta^2 = 0.1$). Bonferroni correction was used to adjust the p-values in the multiple comparisons, with a critical level of significance that $p = 0.05/6$ (Armstrong, 2014). The same adjustment and the critical level of significance were applied in the following ANOVA analysis. Results revealed that the scores in literal praise and literal criticism were significantly higher than in ironic praise and ironic criticism (adjusted $p < 0.001$) and there were no significant differences between the scores in literal praise and literal criticism and between the scores in ironic praise and ironic criticism (adjusted $p > 0.05$).

The interaction between group and story type was also significant (F [3,141] = 4.76, $p < 0.01$, $\eta^2 = 0.1$). Further analysis of simple effects showed that the effects of group on ironic criticism and ironic praise were significant (F [1,192] = 27.20, $p < 0.001$; F [1,192] = 10.16, $p = 0.001$). The irony comprehension of TD group is

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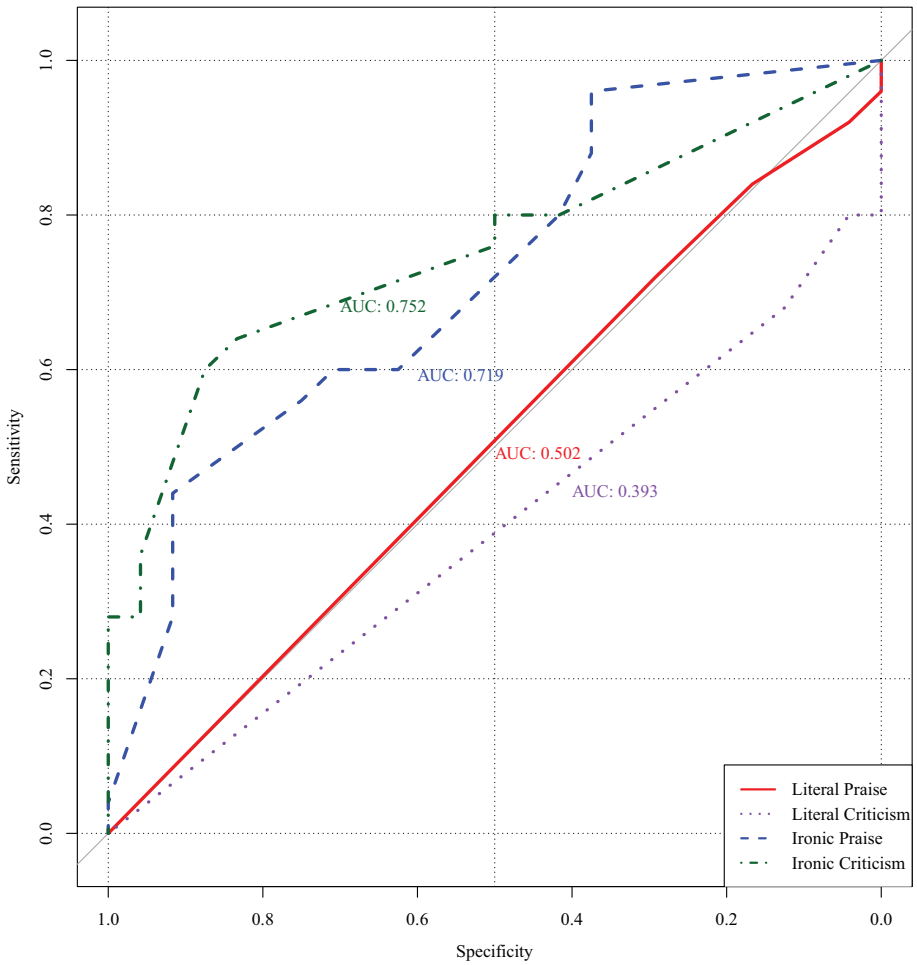

FIGURE 2 Receiver operating characteristics (ROC) curves of scores of literal praise, literal criticism ironic praise, and ironic criticism, on autism spectrum disorder and typically developing children. The area under the curve (AUC) can be ranked from high to low as ironic criticism, ironic praise, literal praise, and literal criticism.

TABLE 6 Correlations of task measures.

	Comprehension of irony		Executive functions			Theory of mind	
	Literal comprehension	Irony comprehension	Inhibition	Flexibility	Working memory	1st ToM	2nd ToM
AQ	−0.19	−0.40*	−0.23	0.03	0.03	−0.47**	−0.42*
Literal comprehension	–	0.03	0.75***	0.28	−0.10	0.38*	0.30
Irony comprehension		–	0.24	−0.07	0.11	0.39*	0.45**
Inhibition			–	0.23	−0.26	0.50**	0.45**
Flexibility				–	−0.25	0.35*	0.29
Working memory					–	−0.24	−0.17
1st ToM						–	0.93***

Note: *$p < 0.05$; **$p < 0.01$; ***$p < 0.001$, two-tailed test, as below.
Abbreviations: 1st ToM, first-order theory of mind; 2nd ToM, second-order theory of mind; AQ, Autism-spectrum Quotient.

significantly better than the irony comprehension of ASD group for both ironic criticism and ironic praise (adjusted $p < 0.001$). However, the effects of group on literal criticism and praise was not significant (F [1,192] = 3.19, $p = 0.08$; F[1,192] = 0.009, $p = 0.92$; see Figure 1).

We performed a two-way repeated ANOVA with the ToM score for the group (ASD and TD groups) as the between-subjects factor, and the ToM task type (1st ToM, 2nd ToM) as the within-subject factor. The main effect of group was significant (F [1,48] = 20.54, $p < 0.001$). Performance on ToM task in TD group was better than the performance in ASD group (adjusted $p < 0.001$). The main effect of ToM task type and the interaction between group and ToM task type were not significant (F [1,48] = 0.14, $p = 0.71$; F [1,48] = 0.38, $p = 0.54$).

We performed a two-way repeated ANOVA with the EF score for the group type (ASD and TD groups) as a

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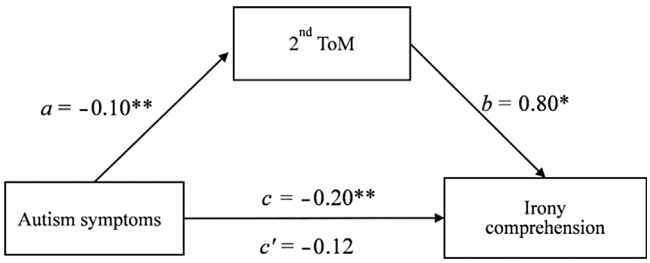

FIGURE 3 Second-order theory of mind 2nd (ToM) as a mediator between autistic traits and irony comprehension. *$p < 0.05$; **$p < 0.01$. (a), b, (c), c', coefficient.

between-subjects factor, and the EF task type (inhibition, flexibility, working memory) as the within-subject factor. Significant differences in EF task performance were found between autistic children and TD controls (F [1,48] = 5.03, $p < 0.05$). Performance on EF task in TD group was better than the performance in ASD group ($p < 0.001$, Bonferroni corrections). The interaction between group and EF task type was not significant (F [2,96] = 1.78, $p = 0.17$).

Discriminant validity of irony story on ASD

Owing to the significant difference in story type between autistic children and TD children, this study examined the possible discriminant effect of irony stories on ASD. Discriminant validity and calculation of the optimal cut-off score were performed using the ROC (Fischer, Bachman, & Jaeschke, 2003), with results presented in Table 5. Test accuracy was measured using the area under the curve (AUC), with an area of 1 representing perfect classification and an area of 0.5 showing a random result.

When differentiating ASD from TD, ironic criticism (ROC–AUC = 75.8% confidence interval [CI] 0.62–0.89) and praise (ROC–AUC = 72.4% CI 0.58–0.86) showed better validity than literal criticism (ROC–AUC = 51.5% CI 0.29–0.53) and praise (ROC–AUC = 40.9% CI 0.38–0.65). Only the two types of irony stories were relatively ideal tools to discriminate between autistic children and TD children, which did not show a significant difference ($Z = -0.51$, $p = 0.61$). Overall, compared with measurements on 4 types of stories (literal stories and ironic stories) the two types of ironic stories showed high specificity and sensitivity for detecting ASD (see Figure 2).

Correlation analysis

Simple Pearson correlation analysis was performed to reveal the relationship between irony comprehension performance and AQ, ToM, and EF scores, adjusted with false discovery rate (FDR; Benjamini & Hochberg, 1995; Table 6). The results showed that performance of irony comprehension was significantly correlated with the 1st ToM level ($r = 0.39$, $p = 0.019$), the 2nd ToM level ($r = 0.45$, $p = 0.006$), and autism symptoms ($r = -0.40$, $p = 0.004$). The performance of literal comprehension did not correlate with autism symptoms ($r = -0.19$, $p = 0.270$). Neither performance on irony comprehension nor autistic traits significantly correlated with performance on the EF.

Regression analyses

To examine the relative contributions of ToM and EF to irony comprehension, a linear stepwise regression analysis was conducted. With performance of irony comprehension as the dependent variable and AQ, 1st ToM, 2nd ToM, inhibition, flexibility, and working memory in the regression equation as predictors, we found that 2nd ToM was a significant predictor of participants' performance on irony comprehension (R^2 change = 0.20, adjusted $R^2 = 0.18$, F [1,48] = 11.94, $p < 0.01$). This indicates that 2nd ToM accounts for 18% of the variance in irony comprehension performance. The AQ accounted for 15% of the variance in irony comprehension performance ($R^2 = 0.17$, adjusted $R^2 = 0.15$, F [1,48] = 11.94, $p < 0.01$). 1st ToM, inhibition, flexibility, and working memory did not account for any variance in irony comprehension. When the AQ, ToM, and EF were entered into the regression equation, the full model with irony as the dependent variable was significant (R^2 change = 0.36, adjusted $R^2 = 0.33$, F [6,43] = 3.96, $p < 0.01$).

Mediation analyses

Owing to the nonsignificant correlation between EF and irony comprehension, EF was excluded from the mediation model. Model 4 in the SPSS Macro PROCESS (Hayes, 2012) was used to test the mediating role of children's 2nd ToM ability in the relationship between autistic traits and irony comprehension, controlling for children's IQ in all paths. Estimates are presented as bootstrapped standard errors and 95% bootstrapped CIs (Hayes & Scharkow, 2013). The mediation model is shown in Figure 3. The result showed that the effect of AQ on 2nd ToM was −0.10 (SE = 0.03, $t = -3.22$, $p = 0.0024$, 95% CI: −0.16 to −0.04), and the effect of 2nd ToM on irony comprehension was 0.80 (SE = 0.29, $t = 2.78$, $p = 0.0079$, 95% CI: 0.22 to 1.39). Importantly, the effect of AQ on irony comprehension was −0.20 (SE = 0.07, $t = -3.00$, $p = 0.0043$, 95% CI: −0.33 to −0.06). When regarding mediation, the indirect effect of AQ on irony was −0.08, and statistically different from zero (95% CI: −0.16 to −0.01). The direct effect was −0.12 (SE = 0.07, $t = -1.71$, $p = 0.09$, 95% CI: −0.25 to

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TABLE 7 The results of mediation analysis.

Model	Independent variable	Dependent variable	β	R^2	F	df
$Y = cX$	AQ	Irony comprehension	−0.20**	0.19	5.38*	46
$M = aX$	AQ	2nd ToM	−0.10**	0.20	5.81*	46
$Y = aX + bM + c'$	AQ	Irony comprehension	−0.12	0.31	6.69***	45
	2nd ToM		0.80*			

Note: *$p < 0.05$, **$p < 0.01$, ***$p < 0.001$, two-tailed test, as below.
Abbreviations: 1st ToM, first-order theory of mind; 2nd ToM, second-order theory of mind; AQ, Autism-spectrum Quotient.

0.02), which contained 0 at 95% CI. Thus, the direct effect was insignificant and 2nd ToM may have played a complete mediation role in this model, which explained 30.83% of the variance in irony comprehension ($F = 6.69$, $p = 0.0008$). Overall, these results suggested that 2nd ToM completely mediates the relationship between autistic traits and comprehension of irony stories (Table 7).

DISCUSSION

This study examined children's understanding of literal and ironic stories. While autistic children showed intact performance in literal understanding (i.e., literal praise and criticism), they were more fragile in irony comprehension (i.e., irony praise and criticism) compared with the TD group. These findings provide further evidence supporting a deficit in irony comprehension in autistic children, aligning with many previous studies (e.g., Deliens, Antoniou, et al., 2018; Deliens, Papastamou, et al., 2018; Gernsbacher & Pripas-Kapit, 2012; Panzeri et al., 2022). ROC analysis indicated that irony comprehension in ASD showed high specificity and sensitivity for detecting ASD. Furthermore, irony comprehension was significantly correlated with the level of autistic symptoms and ToM, and ToM mediates the relationship between ASD symptoms and irony comprehension.

Atypical irony comprehension in autistic children

As a neurodevelopmental disorder, the clinical symptoms of ASD are believed to result from limited cognitive deficits (Frith, 2012). Currently, there are no diagnostic cognitive markers for ASD (Mandy et al., 2014). The existing criteria for diagnosis and classification of ASD are mostly based on behavioral descriptions of clinical experiences, observation in familiar settings, and parental interviews (Carruthers et al., 2021; Kim et al., 2019; Le Couteur et al., 1989; Lord et al., 1989; Rutter, 1988; Volkmar et al., 1987; Wong et al., 2004), which lacks cognitive basis. Because an AUC between 0.7 and 0.9 indicates moderate accuracy (Akobeng, 2007), the ROC analysis in this study showed that performance on irony comprehension was moderately better at detecting autistic children with high specificity and sensitivity. Thus, this study identified a remarkable cognitive marker for autistic children, contributing to better diagnosis and understanding of ASD.

Additionally, previous research on other clinical conditions, such as schizophrenia patients (Ziv et al., 2011), attention-deficit hyperactivity disorder (Caillies et al., 2014), and patients with traumatic brain injury (Dennis et al., 2013), also revealed the impairments on irony comprehension. These results suggest that atypical irony comprehension in autistic children may not be a maker solely associated with ASD and deficit in some other fundamental abilities (e.g., ToM) might account for the poor irony comprehension rather than the autistic trait itself.

The role of 2nd ToM in irony comprehension

The results showed that children with 2nd ToM could better understand the attitudes and meanings of irony speakers. This is consistent with the hypothesis proposed by Happé (1993) that the ability to understand irony was linked to 2nd ToM. One possible reason is that identifying pragmatic insincerity is a prerequisite for understanding irony (Creusere, 2000; Kumon-Nakamura et al., 1995; Winner & Leekam, 1991), and ToM provides a more detailed description of meta-representational reasoning in pragmatic insincerity recognition (Gai et al., 2003). Pragmatic insincerity is thought to be one of the main features of irony, which entails a violation of the common communicative function of speech (Kumon-Nakamura et al., 1995; Wilson, 2013). Kumon-Nakamura et al. (1995) suggested that clarifying pragmatic insincerity could help people pay attention to the intentions behind and recognize the speaker's attitude conveyed by ironic speech (Camp, 2012; Creusere, 2000).

Intention attribution is a component of 2nd ToM and enables individuals to identify pragmatic insincerity and the reason for the speaker's intention in the irony context (Sullivan et al., 1994; Sullivan et al., 1995; Wilson, 2013). Other components of 2nd ToM can provide more contextual information for inferences of irony (Filippova & Astington, 2010). For example, in the following ironic story, "Dad is sick in bed, and Lanlan goes

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to the drugstore to buy medicine. After taking the medication, Dad's condition gradually improved. Dad asked her, 'How do you spend money so recklessly?'" children need to make several inferences to understand this ironic story successfully. They must infer that Lanlan knew that Dad knew what she did (second-order knowledge inference) and why she did it (belief inference) and that Dad did not want her to believe what he had just said (intention inference). Because children with higher levels of 2nd ToM can better conduct meta-representational reasoning on irony stories and more accurately identify pragmatic insincerity, they have a better understanding of the attitudes and meanings of irony speakers.

Although various abilities might affect the performance in understanding irony, this study and many previous studies suggest that ToM is crucial. For example, previous studies have already shown the relationship between 2nd ToM and irony understanding in clinical groups such as attention deficit hyperactivity disorder (Caillies et al., 2014), schizophrenia (Bosco & Gabbatore, 2017; Mo et al., 2008), traumatic brain injury (Cordonier et al., 2020), and Parkinson's disease (Monetta et al., 2009). They demonstrated that when participants have impairments in the 2nd ToM tasks, they usually behave worse in irony comprehension tasks.

In addition, various neurological evidence also indicated that impairments in irony understanding were related to greater activation in the area associated with intention inferring (e.g., medial prefrontal cortex, temporal sulcus, and parietal regions; Schnell et al., 2016; Shibata et al., 2010; Spotorno et al., 2012). Meanwhile, autistic children showed abnormally low activation in prefrontal and parietal areas (Baron-Cohen, 2004). Given the structural impairments, even with some other comorbidities, autistic children might still have worse performance in irony understanding.

Atypical irony comprehension might not be a marker exclusively associated with the condition of ASD. However, considering understanding irony is associated with TOM and the deficit in ToM is one of the most influential cognitive theories of ASD (Rajendran & Mitchell, 2007), it can be reasonably proposed that understanding irony can serve as an important referable cognitive marker for ASD.

The role of EF in irony comprehension

Previous studies have shown that deficits in EF result in rigid thinking, impulsivity, disinhibition, and poor response regulation (McDonald et al., 2014). Poor inhibition might weaken autistic individuals' monitoring over their attention, thoughts, and behaviors, leading to an inability to inhibit inappropriate responses to contextual demands in tasks (Eigsti et al., 2011). Furthermore, impairments in working memory and cognitive flexibility make it difficult for autistic children to constantly manipulate information and shift from literal to intended meanings (Cardillo et al., 2020; Diamond, 2013; Giofrè et al., 2013). However, although the inhibitory component was deficient in autistic children, the relationship between EF and irony comprehension was not significant in this study.

One explanation for this might be the distinct roles of the different components of EF in social cognition. Based on the differences in the prefrontal function of the brain, EF can be divided into "cold" and "hot" components (Salehinejad et al., 2021). "Cold" EF mainly involves the functions of dorsolateral prefrontal and other brain regions, which mainly include relatively abstract and decentralized pure cognitive processing, including working memory, inhibitory control, cognitive flexibility, and other subcomponents (Zelazo & Carlson, 2012). "Hot" EF refers to cognitive processing under high emotional/motivated involvement, including delayed gratification and emotional decision-making, which is related to the ventromedial prefrontal and orbital frontal areas (Antonini et al., 2015; Crone & Steinbeis, 2017). Defective "cold" EF leads to the failure of goal orientation and attention control, specifically in the aspects of attention continuity, selection, distribution, and so forth. Defects in "hot" EF lead to failure of motivation processing in the emotional or social context (Zelazo, 2020). Irony comprehension is a social cognition task; thus, it might mainly rely on "hot" EF, which was not assessed in this study. The EF tasks employed in this study are "cold" EF tasks involving purely cognitive information processing; therefore, the association between EF and irony comprehension (i.e., a process of social cognition) is not as strong as that between ToM and irony comprehension. Jones et al. (2018) used a structural equation model to explore the relationship between ToM, EF, and ASD symptoms and consistently found that ToM, rather than EF, was directly associated with social communication symptoms, which is consistent with our finding that ToM, rather than EF, is more important in irony comprehension.

Limitations and future directions

This study had some limitations that should be addressed. The first limitation is about participants. Given the small sample size, average IQ, and the exclusion of types of comorbidities, these results should be interpreted with caution. Especially, in the clinical world, cooccurring conditions can be commonly observed, and the comorbid conditions may affect the findings observed in pure group of ASDs. For example, research has revealed the relationship between irony understanding and ADHD and schizophrenia which are common comorbidities of autistic children (Gargaro et al., 2011; Hommer & Swedo, 2015). In addition, although ASD is diagnosed four times more often in males than in females

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(DSM-5), some additional autism spectrum condition symptoms displayed different patterns of sex differences in autism spectrum conditions and TD groups, including measures of EF, empathizing and systemizing traits, and play behaviors (Hull et al., 2017). This finding suggests that research on irony comprehension should also consider sex differences. Furthermore, gender identity and biological sex are two distinct terms. Gender identity is a complex issue when considering the ASD group (van Schalkwyk et al., 2015) and it also might be a sensitive question to ask the gender identity in China. Thus, we just recorded the biological sex of participants. However, aside from biological sex recorded in this study, the gender identity of participants might be also a potential factor that might have an impact on autistic children's social behaviors (Abelson, 1981). These findings suggest that research on irony comprehension should also consider the effect of gender identity. Since various abilities may affect understanding irony, future research should be extended to a wider age range, and larger samples including more participants with comorbidity, participants with lower IQ, and sex-balance to draw a more general conclusion.

Second, better performance on 2nd ToM tasks not only represents a higher level of ToM, but also indicates adequate language capacity to assimilate second-order intentions, beliefs, and knowledge. This kind of language ability can also enable children to settle ironic stories, which usually convey a potential contrast between expectancy and reality (Colston & O'Brien, 2000). Thus, to reveal the relationship between ToM and irony comprehension in autistic children, language should be controlled in future research.

The last limitation is how irony tasks were conducted. Based on previous research, multimodal cues were found influential for verbal irony comprehension. Intonation, facial expression, and prosody all might serve as cues for verbal irony understanding (Attardo et al., 2003; Deliens, Antoniou, et al., 2018; Deliens, Papastamou, et al., 2018; Kreuz & Roberts, 1995). Thus, facial expressions and other contextual cues that would normally happen in a social interaction might help autistic individuals decipher irony. However, in this study, the contextual cues were excluded and children lacked related facial and vocal information input to process irony. It is not clear whether autistic children do not understand irony, simply because they rely on more information input in order to decipher whether irony as a cognitive task or not. Future studies could adopt different forms of irony tasks by comparing the effect of contextual cues on irony comprehension.

CONCLUSION

In summary, the results of the irony comprehension task showed a significant difference in the ability to comprehend irony (ironic praise and criticism) between the ASD and TD groups. Irony comprehension was significantly correlated with 2nd ToM but was not significantly correlated with the 1st ToM task or any component of EF. Furthermore, performance on the EF task was not significantly associated with autistic traits, and 2nd ToM completely mediated the relationship between autistic traits and the comprehension of irony stories.

These findings suggest that irony comprehension may offer a potential behavioral marker for screening autistic children and 2nd ToM may play an important role in the theoretical mechanism underlying the deficit in irony comprehension in autistic children.

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CONFLICT OF INTEREST STATEMENT

There are no conflicts of interest to declare.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

ETHICS STATEMENT

The study was approved by the Institutional Review Board of East China Normal University (HR 220-2019).

ORCID

Yongning Song https://orcid.org/0000-0002-5162-8557
Ziyun Nie https://orcid.org/0009-0007-7105-3926

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